

KUOPIO MAANINKA, Finland

WMO: 027880

Lat: 63.143N

Lon: 27.313E

Elev: 91

StdP: 100.23

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-23.9	-30.4	0.2	-27.7	-26.3	0.3	-23.9	9.1	-2.0	8.2	-2.8	1.0	330	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	27.0	18.8	24.9	17.8	23.1	16.7	20.1	25.0	18.9	23.2	17.7	21.5	2.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	22.1	17.2	12.4	21.1	16.0	11.5	19.9	58.0	24.9	54.2	23.0	50.2	21.6	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.5	-29.4	28.8	6.0	3.1	-33.7	31.0	-37.2	32.8	-40.6	34.5	-45.0	36.7		
(5)			-28.9	21.2	6.1	2.0	-33.2	22.7	-36.8	23.8	-40.2	25.0	-44.6	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-9.3	-7.7	-3.7	2.5	9.7	13.9	17.0	15.1	10.3	3.8	-0.3	-5.0		
	DBStd	10.11	7.28	7.43	5.11	3.60	4.54	3.90	3.45	3.09	3.30	3.56	4.30	6.68		
	HDD10.0	2821	599	494	424	224	61	9	1	2	37	194	310	465		
	HDD18.3	5301	857	728	683	474	268	142	66	108	241	451	560	723		
	CDD10.0	603	0	0	0	1	53	126	218	159	46	1	0	0		
	CDD18.3	42	0	0	0	0	1	8	26	7	0	0	0	0		
	CDH23.3	368	0	0	0	0	28	71	216	53	0	0	0	0		
	CDH26.7	68	0	0	0	0	1	10	44	12	0	0	0	0		
(14) Wind	WSAvg	3.1	2.8	3.1	3.3	3.0	3.2	3.2	2.7	2.7	3.0	3.5	3.2	3.2		
(15) Precipitation	PrecAvg	638	44	38	33	32	48	70	88	70	54	55	52	52		
	PrecMax	811	80	74	71	58	123	122	156	149	99	99	115	86		
	PrecMin	456	12	3	5	9	12	20	22	24	20	12	7	18		
	PrecStd	84	18	18	14	13	27	27	34	25	25	23	23	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	4.9	8.2	16.6	26.4	28.1	30.2	28.4	21.0	13.2	9.7	5.4		
		MCWB	2.5	2.9	3.9	9.3	15.3	17.9	20.3	20.5	16.3	10.9	8.5	4.8		
	2%	DB	1.7	3.1	5.6	13.2	23.6	25.4	27.7	24.3	18.5	11.4	7.2	3.4		
		MCWB	1.1	1.7	2.6	7.9	14.4	17.5	19.7	17.6	14.4	9.9	6.4	2.6		
	5%	DB	0.7	1.6	3.7	10.6	21.0	23.1	25.7	22.4	16.7	10.0	5.9	2.3		
		MCWB	0.3	1.0	1.6	6.1	13.9	16.2	18.8	16.7	13.6	8.8	5.4	1.6		
	10%	DB	-0.1	0.7	2.5	8.5	18.0	21.0	23.7	20.6	15.3	8.7	4.5	1.3		
		MCWB	-0.4	0.2	0.9	4.8	11.5	15.1	17.8	16.0	12.9	7.6	4.0	0.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	3.1	4.4	10.5	16.9	20.0	22.0	21.1	16.4	11.6	8.6	5.0		
		MCDB	3.5	4.4	6.8	14.4	23.6	24.6	26.9	27.0	19.8	12.6	9.3	5.5		
	2%	WB	1.3	2.0	3.0	8.4	15.5	18.6	20.6	18.8	15.2	10.4	6.5	2.6		
		MCDB	1.6	2.8	5.1	12.6	21.6	23.3	25.7	22.3	17.3	11.1	7.0	3.1		
	5%	WB	0.5	1.1	2.0	6.5	14.2	16.9	19.7	17.5	14.2	8.9	5.4	1.7		
		MCDB	0.8	1.5	3.3	9.8	20.6	22.3	24.3	20.6	16.0	9.7	5.8	2.1		
	10%	WB	-0.4	0.3	1.2	5.1	12.3	15.6	18.7	16.7	13.3	7.6	4.1	0.9		
		MCDB	0.0	0.8	2.2	7.9	17.3	20.1	22.9	19.8	15.0	8.6	4.4	1.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.2	7.7	8.3	9.6	8.7	8.9	8.1	6.6	4.6	3.4	4.3		
		MCDBR	3.7	4.5	7.6	12.2	14.5	12.3	12.1	11.3	9.0	6.1	4.4	3.5		
	5% WB	MCWBR	3.6	4.0	5.6	7.6	7.0	5.8	5.7	5.7	5.7	4.8	4.2	3.3		
		MCWBR	3.6	4.4	6.5	11.2	13.4	11.4	10.9	9.2	7.9	5.6	4.3	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.284	0.312	0.350	0.347	0.346	0.370	0.352	0.329	0.311	0.244	N/A		
		taud	2.000	2.111	2.182	2.321	2.456	2.483	2.430	2.467	2.521	2.391	2.090	N/A		
	Ebn at Noon		387	635	777	830	870	877	840	825	776	624	364	N/A		
		Edh at Noon	32	66	92	101	96	96	99	87	68	52	29	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.18	0.74	2.25	3.62	4.78	5.25	5.00	3.82	2.13	0.82	0.20	0.08		
	RadStd		0.03	0.17	0.26	0.44	0.52	0.35	0.52	0.42	0.24	0.14	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	-172	N/A	N/A	N/A

Nomenclature: See separate page